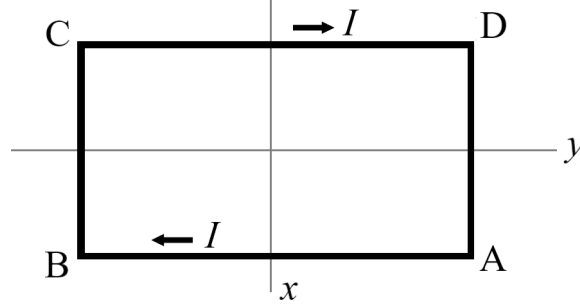




Q1: [40] A rectangular loop carries a current I [A] as shown in Figure 1 (below). Using Biot-Savart's law, calculate the magnetic field at the corner A of the loop. Width of the loop is $2b$ (along y -axis), and the height is $2a$ (along x -axis).



(a) Showing all steps (along edges AB, BC, CD, and DA), obtain $\mathbf{H}_{AB}$, $\mathbf{H}_{BC}$, $\mathbf{H}_{CD}$, and $\mathbf{H}_{DA}$ considering the field point is at A.

(b) Show that $\mathbf{H}_{\text{total}} = \mathbf{H}_{AB} + \mathbf{H}_{BC} + \mathbf{H}_{CD} + \mathbf{H}_{DA} = -\frac{I \sqrt{a^2 + b^2}}{8\pi ab} \mathbf{a}_z \quad \left[\frac{\text{A}}{\text{m}} \right]$

Use the following formula while specifying all involved variables in your calculations for each of the four edges:

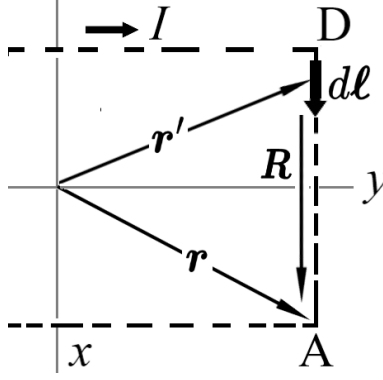
$$d\mathbf{H} = \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{4\pi \|\mathbf{r} - \mathbf{r}'\|^3}$$

Hint: An illustration is provided for edge DA as follows: Provide similar illustrations for all edges in your work.

Solution: Rectangular loop centered at the origin, width $2b$ along the y -axis, height $2a$ along the x -axis. Current I flows **clockwise** (viewed from $+z$). Corners are: $A = (a, b, 0)$, $B = (a, -b, 0)$, $C = (-a, -b, 0)$, $D = (-a, b, 0)$. Field point $\mathbf{r} = (a, b, 0)$ (corner A). Biot-Savart law states:

$$d\mathbf{H} = \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{4\pi \|\mathbf{r} - \mathbf{r}'\|^3},$$

where $\mathbf{r}'$ is the source point on the wire, $d\boldsymbol{\ell}$ is in the direction of current.



Contribution of side AB

On AB : $x' = a$, y' varies from b to $-b$, $z' = 0$. Current direction: $A \rightarrow B$, i.e., $\mathbf{a}_y$ (the minus sign may be incorporated in the limits of integration). $\mathbf{r} = (a, b, 0)$ and $\mathbf{r}' = (a, y', 0)$.

$$d\boldsymbol{\ell} = (0, dy', 0).$$

$$\mathbf{r} - \mathbf{r}' = (a - a, b - y', 0) = (0, b - y', 0).$$

Here $d\boldsymbol{\ell}$ is parallel to $\mathbf{r} - \mathbf{r}'$, so $d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}') = \mathbf{0}$.

$$\boxed{\mathbf{H}_{AB} = \mathbf{0}}.$$

Contribution of side BC

On BC : $y' = -b$, x' varies from a to $-a$, $z' = 0$. Current direction: $B \rightarrow C$, i.e., $-\mathbf{a}_x$. $\mathbf{r} = (a, b, 0)$ and $\mathbf{r}' = (x', -b, 0)$. Then

$$d\boldsymbol{\ell} = (dx', 0, 0).$$

$$\mathbf{r} - \mathbf{r}' = (a - x', b - (-b), 0) = (a - x', 2b, 0).$$

$$d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}') = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ dx' & 0 & 0 \\ a - x' & 2b & 0 \end{vmatrix} = (0, 0, 2b dx').$$

$$\|\mathbf{r} - \mathbf{r}'\|^3 = (s^2 + 4b^2)^{3/2}.$$

Thus

$$d\mathbf{H}_{BC} = \frac{I}{4\pi} \cdot \frac{(0, 0, 2b dx')}{((a - x')^2 + 4b^2)^{3/2}}.$$

Integrating x' from a to $-a$:

$$\mathbf{H}_{BC} = \left(0, 0, \frac{Ib}{2\pi} \int_a^{-a} \frac{dx'}{((a-x')^2 + 4b^2)^{3/2}} \right).$$

Standard integral:

$$\int \frac{dx'}{((x'-a)^2 + k^2)^{3/2}} = \frac{(x'-a)}{k^2 \sqrt{(x'-a)^2 + k^2}}.$$

Here $k = 2b$:

$$\int_a^{-a} \frac{dx'}{((x'-a)^2 + 4b^2)^{3/2}} = \left[\frac{(x'-a)}{4b^2 \sqrt{(x'-a)^2 + 4b^2}} \right]_a^{-a} = \frac{-2a}{4b^2 \sqrt{4a^2 + 4b^2}}.$$

Hence

$$\mathbf{H}_{BC} = \left(0, 0, -\frac{Ib}{2\pi} \cdot \frac{a}{4b^2 \sqrt{a^2 + b^2}} \right) = -\frac{Ia}{8\pi b \sqrt{a^2 + b^2}} \mathbf{a}_z.$$

Contribution of side CD

On CD : $x' = -a$, y' varies from $-b$ to b , $z' = 0$. Current direction: $C \rightarrow D$, i.e., $+\mathbf{a}_y$. Let y' be distance from C : $\mathbf{r}' = (-a, y', 0)$, $y' \in [-b, b]$. Then

$$d\mathbf{l} = (0, dy', 0).$$

$$\mathbf{r} - \mathbf{r}' = (a - (-a), b - y', 0) = (2a, b - y', 0).$$

$$d\mathbf{l} \times (\mathbf{r} - \mathbf{r}') = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & dy' & 0 \\ 2a & b - y' & 0 \end{vmatrix} = (0, 0, -2a dy').$$

$$\|\mathbf{r} - \mathbf{r}'\|^3 = (4a^2 + (b - y')^2)^{3/2} = (4a^2 + (y' - b)^2)^{3/2}.$$

Thus

$$d\mathbf{H}_{CD} = \frac{I}{4\pi} \cdot \frac{(0, 0, -2a dy')}{[4a^2 + (y' - b)^2]^{3/2}}.$$

Integrate t from $-b$ to b :

$$\mathbf{H}_{CD} = \left(0, 0, -\frac{Ia}{2\pi} \int_{-b}^b \frac{dy'}{[4a^2 + (y' - b)^2]^{3/2}} \right).$$

Similar to the above arguments, we obtain

$$\mathbf{H}_{CD} = \left(0, 0, -\frac{Ia}{2\pi} \cdot \frac{b}{4a^2 \sqrt{a^2 + b^2}} \right) = -\frac{Ib}{8\pi a \sqrt{a^2 + b^2}} \mathbf{a}_z.$$

Contribution of side DA

On DA : $y' = b$, x' varies from $-a$ to a , $z' = 0$. Current direction: $D \rightarrow A$, i.e., $+\mathbf{a}_x$. $d\boldsymbol{\ell} = (dx', 0, 0)$, $\mathbf{r} - \mathbf{r}' = (a - x', b - b, 0) = (a - x', 0, 0)$. These are parallel, so $d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}') = \mathbf{0}$.

$$\boxed{\mathbf{H}_{DA} = \mathbf{0}}.$$

Total magnetic field from direct integration

$$\mathbf{H}_{\text{total}} = \mathbf{H}_{AB} + \mathbf{H}_{BC} + \mathbf{H}_{CD} + \mathbf{H}_{DA}.$$

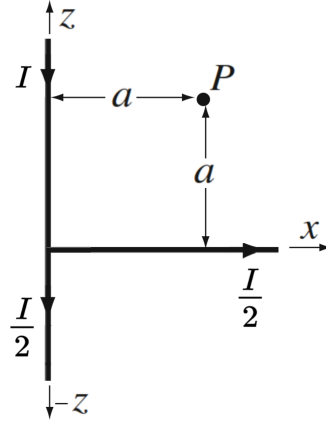
Only $\mathbf{H}_{BC}$ and $\mathbf{H}_{CD}$ contribute, both in $-\mathbf{a}_z$ direction:

$$\begin{aligned} \mathbf{H}_{\text{total}} &= -\frac{Ia}{8\pi b\sqrt{a^2 + b^2}} \mathbf{a}_z - \frac{Ib}{8\pi a\sqrt{a^2 + b^2}} \mathbf{a}_z = -\frac{I}{8\pi\sqrt{a^2 + b^2}} \left(\frac{a}{b} + \frac{b}{a} \right) \mathbf{a}_z. \\ &= -\frac{I}{8\pi\sqrt{a^2 + b^2}} \cdot \frac{a^2 + b^2}{ab} \mathbf{a}_z = -\frac{I\sqrt{a^2 + b^2}}{8\pi ab} \mathbf{a}_z. \end{aligned}$$

Q2: [30] An infinitely long wire carrying a current I [A] splits into two branches, each carrying $0.5 I$ [A] as shown in Figure 2. Determine the magnetic flux density at point $P(a, 0, a)$ using the Biot-Savart's law:

$$d\mathbf{B} = \mu_0 \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{4\pi \|\mathbf{r} - \mathbf{r}'\|^3}$$

Use this formula while specifying all involved variables.



Solution: We have an infinitely long wire carrying current I . Along positive z -axis from $z = \infty$ to $z = 0$ with current flowing downward:

$$d\boldsymbol{\ell}_1 = \mathbf{a}_z dz', \quad \mathbf{r}' = (0, 0, z'), \quad z' \in [0, \infty), \quad \mathbf{r} = (a, 0, a)$$

(physically, current flows from $z' = \infty$ to $z' = 0$). Along positive x -axis from $x = 0$ to $x = \infty$ with current flowing outward:

$$d\boldsymbol{\ell}_2 = \mathbf{a}_x dx', \quad \mathbf{r}' = (x', 0, 0), \quad x' \in [0, \infty), \quad \mathbf{r} = (a, 0, a)$$

Biot–Savart law:

$$\mathbf{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3}.$$

Contribution from the vertical segment: For the vertical wire, $d\boldsymbol{\ell}_1 = \mathbf{a}_z dz'$, with z' integrated from ∞ to 0.

$$\mathbf{r} - \mathbf{r}' = (a, 0, a) - (0, 0, z') = (a, 0, a - z').$$

$$|\mathbf{r} - \mathbf{r}'| = \sqrt{a^2 + (a - z')^2}.$$

$$d\boldsymbol{\ell}_1 \times (\mathbf{r} - \mathbf{r}') = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 0 & dz' \\ a & 0 & a - z' \end{vmatrix} = -\mathbf{a}_y [0 \cdot (a - z') - (dz') \cdot a] = \mathbf{a}_y a dz'.$$

Thus:

$$d\mathbf{B}_1 = \frac{\mu_0 I}{4\pi} \frac{(0, a dz', 0)}{[a^2 + (a - z')^2]^{3/2}}.$$

$$\mathbf{B}_1 = \mathbf{a}_y \frac{\mu_0 I a}{4\pi} \int_{\infty}^0 \frac{dz'}{[a^2 + (a - z')^2]^{3/2}} = -\mathbf{a}_y \frac{\mu_0 I a}{4\pi} \int_0^{\infty} \frac{dz'}{[a^2 + (z' - a)^2]^{3/2}}.$$

$$\mathbf{B}_1 = -\mathbf{a}_y \frac{\mu_0 I a}{4\pi} \frac{(z' - a)}{a^2 \sqrt{a^2 + (z' - a)^2}} \Big|_0^{\infty} = -\mathbf{a}_y \frac{\mu_0 I}{4\pi a} \left(1 + \frac{1}{\sqrt{2}}\right)$$

We can easily prove that the contribution from the horizontal segment also gives the same result as shown below (you may easily prove it):

$$\mathbf{B}_2 = -\mathbf{a}_y \frac{\mu_0 I}{4\pi a} \left(1 + \frac{1}{\sqrt{2}}\right).$$

Total magnetic field:

$$\begin{aligned} \mathbf{B}_{\text{total}} = \mathbf{B}_1 + \mathbf{B}_2 &= -\mathbf{a}_y \frac{\mu_0 I}{4\pi a} \left(1 + \frac{1}{\sqrt{2}}\right) - \mathbf{a}_y \frac{\mu_0 I}{4\pi a} \left(1 + \frac{1}{\sqrt{2}}\right) \\ &= -\mathbf{a}_y \frac{\mu_0 I}{2\pi a} \left(1 + \frac{1}{\sqrt{2}}\right). \end{aligned}$$

The field points in the negative y -direction. This is the magnetic flux density at point $P(a, 0, a)$.

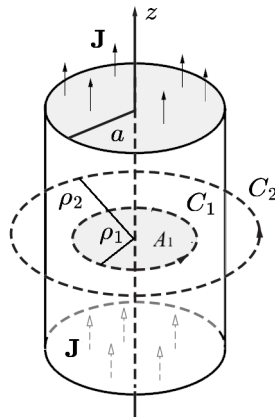
Q3: [30] Consider an infinitely long, straight cylindrical conductor with radius a , carrying a steady current I . The current is non-uniformly distributed such

that the current density $\mathbf{J}$ is proportional to the radial distance ρ from the z -axis. Assume $\mathbf{J} = K(e^\rho - 1) \mathbf{a}_z$ for $\rho \leq a$, where K is a positive constant. This is an obvious fact that $\mathbf{J} = \mathbf{0}$ for $\rho > a$ (outside the conductor).

- (a) [5] Given a and I , what is the value of K ?
- (b) [10] Using Ampere's law, determine the magnetic field intensity $\mathbf{H}$ both **inside** and **outside** the conductor.
- (c) [5] Plot the magnitude of $\mathbf{H}$ to show its variation as a function of ρ .
- (d) [10] Find the curl of $\mathbf{H}$ to establish that

$$\nabla \times \mathbf{H} = \mathbf{J}$$

is true **everywhere**, i.e., both inside and outside the cable.



When using Ampere's law, one can consider the circular Ampere contours C_1 and C_2 with radii ρ_1 and ρ_2 , respectively, where $0 < \rho_1 < a$ and $a < \rho_2 < \infty$, to calculate the magnetic field intensity both inside and outside the conductor.

Solution: Given the current density inside the conductor ($\rho \leq a$):

$$\mathbf{J}(\rho) = K(e^\rho - 1) \mathbf{a}_z, \quad K > 0,$$

and zero outside. Total current I is obtained by integrating $\mathbf{J}$ over the cross-section:

$$I = \iint_S \mathbf{J} \cdot d\mathbf{S}.$$

In cylindrical coordinates, $d\mathbf{S} = \rho d\rho d\phi \mathbf{a}_z$, so

$$I = \int_{\phi=0}^{2\pi} \int_{\rho=0}^a K(e^\rho - 1) \rho d\rho d\phi.$$

Since the integrand is independent of ϕ ,

$$I = 2\pi K \int_0^a (e^\rho - 1) \rho d\rho.$$

Now evaluate the integral:

$$\int_0^a \rho e^\rho d\rho = [e^\rho(\rho - 1)]_0^a = e^a(a - 1) - e^0(0 - 1) = e^a(a - 1) + 1.$$

$$\int_0^a \rho d\rho = \frac{a^2}{2}.$$

Thus

$$I = 2\pi K \left[e^a(a - 1) + 1 - \frac{a^2}{2} \right].$$

Hence

$$K = \frac{I}{2\pi \left[e^a(a - 1) + 1 - \frac{a^2}{2} \right]}.$$

Part (b): Cylindrical symmetry implies $\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$. Take a circular Ampèrian loop C of radius ρ centered on the z -axis:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}(\rho).$$

Since $d\mathbf{l} = \rho d\phi \mathbf{a}_\phi$,

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = H_\phi(\rho) \cdot 2\pi\rho.$$

So

$$2\pi\rho H_\phi(\rho) = I_{\text{enc}}(\rho).$$

Inside the conductor ($0 < \rho \leq a$)

$$I_{\text{enc}}(\rho) = \int_0^\rho \int_0^{2\pi} J_z(\rho') \rho' d\phi d\rho' = 2\pi K \int_0^\rho \rho' (e^{\rho'} - 1) d\rho'.$$

We already computed a similar integral:

$$\int_0^\rho \rho' e^{\rho'} d\rho' = e^\rho(\rho - 1) + 1,$$

$$\int_0^\rho \rho' d\rho' = \frac{\rho^2}{2}.$$

Thus

$$I_{\text{enc}}(\rho) = 2\pi K \left[e^\rho(\rho - 1) + 1 - \frac{\rho^2}{2} \right].$$

Therefore

$$\mathbf{H}_{\text{inside}} = \frac{K}{\rho} \left[e^\rho(\rho - 1) + 1 - \frac{\rho^2}{2} \right] \mathbf{a}_\phi, \quad 0 < \rho \leq a$$

Outside the conductor ($\rho \geq a$). For $\rho \geq a$, all current I is enclosed:

$$I_{\text{enc}}(\rho) = I.$$

Thus $H_\phi(\rho) = \frac{I}{2\pi\rho}$.

$$\mathbf{H}_{\text{outside}} = \frac{I}{2\pi\rho} \mathbf{a}_\phi, \quad \rho \geq a.$$

Part (d) – Verify $\nabla \times \mathbf{H} = \mathbf{J}$ everywhere

In cylindrical coordinates, for $\mathbf{H} = H_\phi(\rho) \mathbf{a}_\phi$ (with $H_\rho = H_z = 0$ and $\partial/\partial\phi = \partial/\partial z = 0$):

$$\nabla \times \mathbf{H} = \frac{1}{\rho} \frac{d}{d\rho} [\rho H_\phi(\rho)] \mathbf{a}_z.$$

From Ampère's law: $\rho H_\phi(\rho) = \frac{I_{\text{enc}}(\rho)}{2\pi}$. Thus

$$\frac{1}{\rho} \frac{d}{d\rho} [\rho H_\phi(\rho)] = \frac{1}{\rho} \frac{d}{d\rho} \left[\frac{I_{\text{enc}}(\rho)}{2\pi} \right] = \frac{1}{2\pi\rho} \frac{dI_{\text{enc}}}{d\rho}.$$

How $\frac{dI_{\text{enc}}}{d\rho}$ is the current through an annulus of radius ρ and thickness $d\rho$:

$$\frac{dI_{\text{enc}}}{d\rho} = J_z(\rho) \cdot 2\pi\rho.$$

Substituting:

$$\frac{1}{\rho} \frac{d}{d\rho} [\rho H_\phi(\rho)] = \frac{1}{2\pi\rho} \cdot [J_z(\rho) \cdot 2\pi\rho] = J_z(\rho).$$

Hence

$$\nabla \times \mathbf{H} = J_z(\rho) \mathbf{a}_z = \mathbf{J}(\rho).$$

- Inside ($\rho \leq a$): $J_z(\rho) = K(e^\rho - 1)$. - Outside ($\rho > a$): $J_z(\rho) = 0$, and indeed I_{enc} is constant so derivative zero. Thus the relation holds everywhere.

$$\boxed{\nabla \times \mathbf{H} = \mathbf{J} \quad \text{for all } \rho}$$